TEXORA · LECTURE SHEET

# Electronics Engineering: Diodes, Transistors and Amplifiers

*Semiconductors and the p–n junction diode — its operation, characteristic and applications; half-wave and full-wave rectifiers; the transistor as an amplifier; and the operational amplifier — ideal properties, inverting and non-inverting amplifiers, integrator and differentiator.*

EEE-2203 · BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING · TOPIC 5 · LEVEL 2 TERM 2

## 1. Semiconductors: the basics

A **semiconductor** (silicon, germanium) has a conductivity between that of a conductor and an insulator, and — the useful part — it can be controlled. In a pure (intrinsic) crystal, a few electrons break free of their bonds at room temperature, leaving **holes** (positive vacancies); both carry current. **Doping** adds impurity atoms to create an extrinsic semiconductor: **n-type** material is doped with a pentavalent impurity (phosphorus, arsenic) giving free electrons as majority carriers; **p-type** material is doped with a trivalent impurity (boron, gallium) giving holes as majority carriers. Joining p-type and n-type material in one crystal makes the **p–n junction**, the structure behind nearly every electronic device.

## 2. The p–n junction diode: operation

At the junction, electrons diffuse from n to p and holes from p to n, leaving a thin **depletion region** stripped of free carriers and a built-in **barrier potential** (about 0.7 V for silicon, 0.3 V for germanium) opposing further diffusion.

- **Forward bias** (p-side to +, n-side to −): the applied voltage opposes the barrier; above the knee voltage (0.7 V Si, 0.3 V Ge) the depletion region collapses and current rises steeply — the diode conducts.
- **Reverse bias** (p-side to −): the barrier widens, and only a tiny saturation current (microamperes) flows. If the reverse voltage exceeds the **breakdown voltage**, current avalanches upward and an ordinary diode is destroyed (the Zener diode is designed to work in controlled breakdown as a voltage reference).

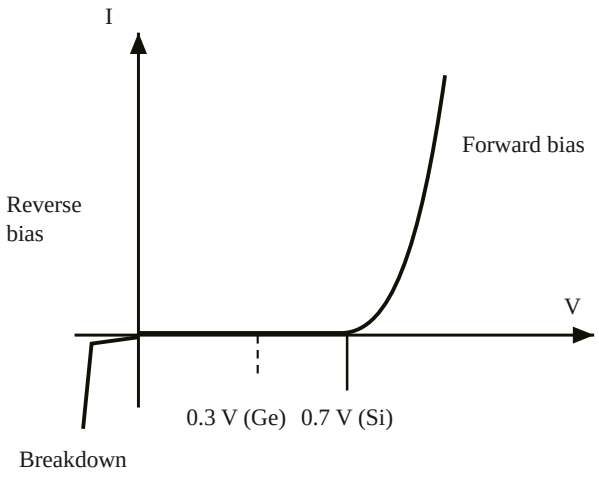



Fig. 1 — The diode V–I characteristic: negligible conduction until the knee, then a steep rise; a small reverse current until breakdown.

**Applications of diodes:** rectification (A.C. to D.C., next section), signal demodulation in radio receivers, clamping and clipping in wave-shaping, voltage regulation (Zener), light emission (the LED — Topic 6), and freewheeling protection across inductive loads.

## 3. Rectifiers: converting A.C. to D.C.

**Half-wave rectifier:** one diode in series with the load passes only the half-cycles of the right polarity; the other half-cycles are blocked. The output is a series of pulses — D.C. in direction but far from smooth. Its average value is $V_{dc} = V_m/\pi \approx 0.318\ V_m$, and its ripple factor is 1.21 (ripple bigger than the D.C.!), so it is used only for low-power, undemanding supplies.

**Full-wave rectifiers** use both half-cycles. The **centre-tap** circuit uses two diodes and a centre-tapped transformer secondary, each diode conducting on alternate half-cycles; the **bridge** uses four diodes and an ordinary transformer, steering both half-cycles through the load in the same direction. The output average doubles: $V_{dc} = 2V_m/\pi \approx 0.636\ V_m$, the ripple frequency doubles (100 Hz from a 50 Hz supply), and the ripple factor falls to 0.48. A reservoir capacitor across the load charges to near the peak and discharges between pulses, smoothing the output further (in the Sessional course you will see rectifiers with and without this filter).

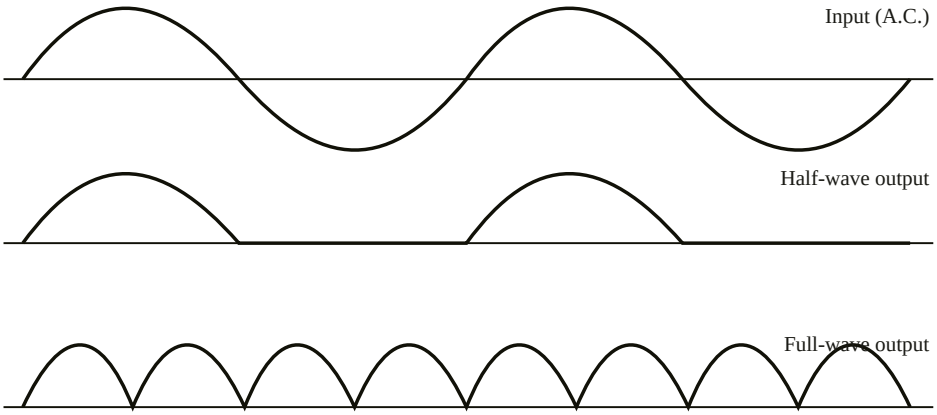



Fig. 2 — Rectifier waveforms: the half-wave rectifier discards half the input; the full-wave rectifier uses both halves, so its pulses come twice as often.

**WORKED EXAMPLE — RECTIFIER AVERAGES**

A rectifier is fed from a secondary giving $V_m$ = 20 V peak. Half-wave: $V_{dc} = 20/\pi$ = **6.37 V**. Full-wave: $V_{dc} = 2 \times 20/\pi$ = **12.73 V**. The full-wave circuit delivers twice the average voltage from the same peak — and with half the ripple factor.

## 4. The transistor (BJT): basics

The **bipolar junction transistor** is a three-layer sandwich — npn or pnp — with terminals **emitter (E)**, **base (B)** and **collector (C)**. The base–emitter junction is forward-biased and the base–collector junction reverse-biased for amplification. A small base current $I_B$ controls a much larger collector current:

$$I_E = I_B + I_C, \quad \beta = I_C/I_B \text{ (current gain, typically 50–300)}$$

Because a small input current commands a large output current, the transistor amplifies — the working principle of audio amplifiers, control circuits and the switching stages inside digital electronics. Worked numbers: $I_B$ = 20 μA with β = 100 gives $I_C = 100 \times 20$ μA = **2 mA** and $I_E$ = 2.02 mA. The amplified, integrated descendant of the transistor amplifier is the operational amplifier.

## 5. The operational amplifier and its ideal properties

An **operational amplifier (op-amp)** is a high-gain D.C. amplifier in a single integrated circuit, with two inputs — inverting (−) and non-inverting (+) — and one output; it amplifies the *difference* between the inputs. The **ideal op-amp** is assumed to have:

| PROPERTY | IDEAL VALUE AND CONSEQUENCE |
|---|---|

| | |
|---|---|
| Open-loop voltage gain | Infinite — so with negative feedback the gain is set entirely by the external resistors |
| Input impedance | Infinite — the inputs draw no current (“no current enters either input”) |
| Output impedance | Zero — the output holds its voltage whatever load is connected |
| Bandwidth | Infinite; offset voltage and drift zero |

With negative feedback, these give the two **golden rules**: (1) the output does whatever is needed to make the two inputs equal — a **virtual earth** at the inverting input when the other input is grounded; (2) no current flows into the inputs. Almost every op-amp circuit is analysed in seconds with these two rules. Real op-amps (such as the classic 741) approach the ideal closely enough at low frequencies, but their output cannot exceed the supply rails — a practical limit called saturation.

## 6. The inverting and non-inverting amplifiers

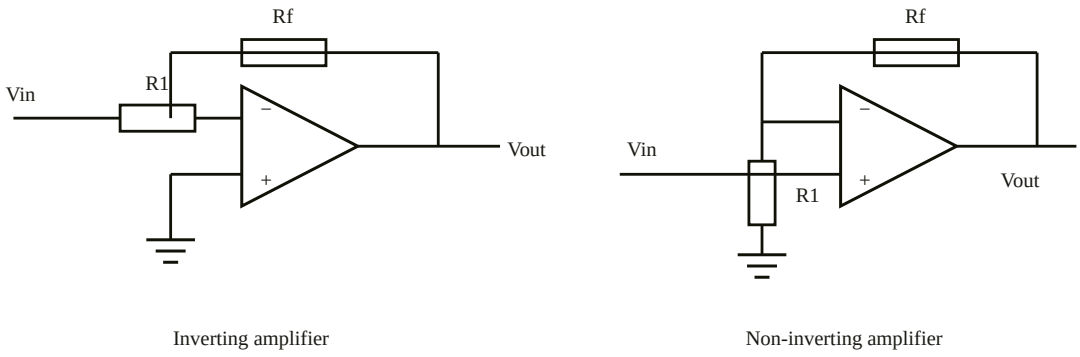



Fig. 3 — The two basic op-amp amplifier circuits. In each, the feedback resistor Rf works against R1 to set the gain.

**Inverting amplifier:** the signal enters through R1 to the inverting input; the non-inverting input is grounded, so the inverting input is a virtual earth. The input current Vin/R1 must flow through Rf (none enters the op-amp), giving

$$V_{out}/V_{in} = -Rf/R1$$

**Non-inverting amplifier:** the signal enters the (+) input; feedback returns a fraction of the output (set by the R1–Rf divider) to the (−) input, which must equal Vin, giving

$$V_{out}/V_{in} = 1 + Rf/R1$$

**WORKED EXAMPLE — OP-AMP GAINS**

> Inverting: R1 = 10 kΩ, Rf = 100 kΩ, Vin = 0.5 V → gain = −100/10 = −10, so $V_{out}$ = **−5 V** (inverted, as the minus sign requires).
> Non-inverting: R1 = 10 kΩ, Rf = 30 kΩ, Vin = 1.2 V → gain = 1 + 30/10 = 4, so $V_{out}$ = **+4.8 V**.

## 7. Integrator and differentiator

Replacing resistors by a capacitor turns the inverting amplifier into a calculus machine. In the **inverting integrator**, Rf is replaced by a capacitor C: the constant input current Vin/R charges the capacitor, so the output is a ramp proportional to the time-integral of the input:

$$\text{Integrator: } V_{out} = -(1/(RC)) \int V_{in}\, dt \quad \cdot \quad \text{Differentiator: } V_{out} = -RC \cdot dV_{in}/dt$$

In the **differentiator**, R1 is replaced by a capacitor: the input current is C·dVin/dt and the output is proportional to the input's rate of change. Example: an integrator with R = 100 kΩ and C = 1 µF (RC = 0.1 s) fed with a steady +2 V input ramps its output at −(2/0.1) = −20 V per second, reaching −2 V after 0.1 s (until it saturates at the supply rail). Square waves integrate to triangles and differentiate to spikes — these two circuits sit inside analogue computers, function generators and the control loops of drives.

## 8. Key terms

- **Doping:** adding controlled impurities to a semiconductor to make it n-type (excess electrons) or p-type (excess holes).
- **Depletion region:** the carrier-free layer at a p–n junction that creates the barrier potential.
- **Knee voltage:** the forward voltage (≈ 0.7 V Si, 0.3 V Ge) at which diode current begins to rise steeply.
- **Ripple factor:** the ratio of the A.C. (ripple) content of a rectifier output to its D.C. value; lower is smoother.
- **Current gain β:** in a transistor, the ratio $I_C/I_B$.
- **Virtual earth:** an op-amp input held at earth potential by feedback, without being connected to earth.
- **Negative feedback:** returning a fraction of the output to the inverting input; it trades the op-amp's huge gain for accuracy and stability.

## 9. Key points

- A diode conducts only when forward-biased beyond its knee voltage; that one-way behaviour is the whole basis of rectification.

- Half-wave: $V_{dc} = 0.318\ V_m$, ripple factor 1.21. Full-wave: $V_{dc} = 0.636\ V_m$, ripple factor 0.48, ripple at twice the supply frequency.
- In a BJT, $I_E = I_B + I_C$ and $I_C = \beta I_B$: a small base current controls a large collector current.
- The ideal op-amp has infinite gain and input impedance and zero output impedance; with negative feedback the golden rules (inputs equal, no input current) solve its circuits.
- Inverting gain $-Rf/R1$; non-inverting gain $1 + Rf/R1$; a capacitor in the feedback path gives an integrator, in the input path a differentiator.

## 10. Summary

Electronics begins with controlled conductivity: doping makes n- and p-type semiconductors, their junction makes the diode — a one-way valve whose V–I curve explains rectification — and three junctions make the transistor, in which a small base current controls a large collector current. Integrated and perfected, the amplifier becomes the op-amp: an almost ideal building block whose behaviour is fixed by a few external components — resistors set the gain of the inverting and non-inverting amplifiers, and capacitors turn it into an integrator or differentiator. These are the devices inside every power supply, sensor interface and control system met elsewhere in this course.

## 11. Practice

Answer in your own words or by calculation; brief answers follow. No multiple-choice questions.

1. A transistor has $\beta = 80$ and a base current of 15 μA. Find the collector and emitter currents.
2. An inverting amplifier has R1 = 5 kΩ and Rf = 50 kΩ. What is the output for an input of −0.2 V?
3. A non-inverting amplifier has R1 = 10 kΩ, Rf = 20 kΩ and an input of 0.8 V. Find the output.
4. A half-wave rectifier is fed with $V_m = 10$ V. Find its average output voltage, and state what a full-wave rectifier would give from the same peak.
5. List the four ideal op-amp properties, and explain in one or two lines why an op-amp is almost always used with negative feedback rather than open-loop.

### Answer key

1. $I_C = \beta I_B = 80 \times 15\ \mu A = 1.2$ mA. $I_E = I_B + I_C = 0.015 + 1.2 = 1.215$ mA.
2. Gain $= -Rf/R1 = -50/5 = -10$. $V_{out} = -10 \times (-0.2) = +2$ V.
3. Gain $= 1 + Rf/R1 = 1 + 20/10 = 3$. $V_{out} = 3 \times 0.8 = 2.4$ V.
4. Half-wave: $V_{dc} = V_m/\pi = 10/\pi = 3.18$ V. Full-wave: $2V_m/\pi = 6.37$ V.

5. Infinite open-loop gain, infinite input impedance, zero output impedance, infinite bandwidth (with zero offset). Open-loop, the infinite gain means any tiny input difference saturates the output, so the device can only act as a comparator; negative feedback reduces the gain to a precise value set by external resistors and makes the circuit stable, linear and predictable.

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